

Bhukya Roopak Krishna

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Professional Summary

AI/ML Engineer who ships production AI systems end-to-end, from model development to deployment. Built **AI-powered computer vision proctoring** and **AI-based KYC verification** systems used in production, **LLM-powered agentic pipelines** for market intelligence, and a **real-time AI voice agent** for primary care. Experienced in **LLM fine-tuning**, **RAG**, **multi-agent systems**, and **MLOps**, owning the backend, AI workflows, and infrastructure from prototype to production.

Education

B.Tech, Materials Science & Metallurgical Engineering

2021 – 2025

Indian Institute of Technology (IIT) Hyderabad

Technical Skills

Languages: Python, C, C++, JavaScript, TypeScript

AI / ML: PyTorch, Scikit-Learn, TensorFlow, PEFT, LoRA, NLP, Computer Vision, LLM Fine-Tuning

LLM / GenAI: RAG, Embeddings, Hybrid Search, Agentic AI, LangChain, LangGraph, Qdrant, FAISS, vLLM

MLOps & Cloud: Kubeflow, Kubeflow Pipelines, MLflow, Docker, GitHub Actions, AWS

Backend / Web: React, Next.js, FastAPI, Flask, REST APIs

Databases & Tools: PostgreSQL, MySQL, Git, GitHub, Jupyter, Postman

Professional Experience

AI Full Stack Developer

Nov 2025 – Jul 2026

Net Connect Global Pvt. Ltd., Bengaluru

- Built an **AI interview proctoring system** that detects suspicious face and voice presence, gaze direction, and mobile devices through computer vision and AI Agents, while also flagging keystroke patterns and unauthorized browser activity during interviews, automatically generating **30+ proctoring reports daily** for review.
- Designed a **KYC verification pipeline** that extracts information from Aadhaar and PAN documents using OCR, with **OpenAI** handling unclear or incomplete fields, and matches document photos against candidate faces, reducing manual identity checks across **1,000+** production verifications.

Machine Learning Intern

Jun 2025 – Sep 2025

Mobius by Gaian, Hyderabad

- Improved **knowledge graph link prediction by 33%** by replacing R-GCN with CompGCN models and deploying a reusable end to end **MLOps workflow** on Kubeflow Pipelines covering GNN training, evaluation, and versioning.
- Fine-tuned **Mistral** and other LLMs on proprietary domain-specific datasets using **PEFT** techniques, achieving a **1.4% accuracy gain** in task performance while reducing inference latency by **0.3 seconds**.

Projects

Clinexa: Agentic Primary-Care Voice Assistant | GitHub

Sep 2026 – Oct 2026

- Engineered a **clinical RAG pipeline** over WHO guidelines using **Qdrant**, **BGE embeddings**, **BM25 with RRF**, and **cross-encoder reranking**, with age/pregnancy filters that keep adult callers out of paediatric content (**110 ms** p50 retrieval).
- Built a **real-time voice layer** (**Twilio Media Streams**, **FastAPI WebSockets**, **Deepgram STT**) and an **evaluation harness** (Hit@k, MRR, NDCG) benchmarking retrieval and reranking accuracy versus latency, with **180+ automated tests**.

Market Intelligence RAG System | GitHub

Aug 2026 – Sep 2026

- Engineered a **multi-agent RAG system** with LangGraph orchestration, LangChain retrieval/tooling, and Qdrant vector search, using Groq for sentiment, event, and market impact analysis across NIFTY 50 companies.
- Architected **9 specialized agents** and **21 tools** to automate news ingestion, document retrieval, event extraction, and contextual synthesis, transforming real time market developments into structured daily intelligence.

Certificate Verification Platform (SkillKendra) | GitHub

Jul 2026 – Aug 2026

- Built a **certificate verification pipeline** with FastAPI that extracts text from uploaded certificates using PaddleOCR, EasyOCR, and Tesseract in parallel, then uses **Selenium** to cross check certificate details against official provider records across **50+ certification providers**.
- Added a **Mistral based AI layer** to normalize inconsistent certificate fields and flag potential fraud, with manual review for cases where the OCR output was unclear or incomplete.

Achievements & Activities

- Contributed to **2 public open-source repositories** focused on **machine learning and AI**, with code fixes and improvements.
- Solved **300+ DSA problems** across LeetCode, GeeksforGeeks, and Codeforces.